Perturbative and Non-Perturbative Wings of the Zeta Building: Kreimer–Connes at $\zeta(s > 0)$ and Vacuum Cosmology at $\zeta(s < 0)$, Unified by the Functional Equation

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Abstract

The Kreimer–Connes Hopf algebra framework established that perturbative QFT renormalization produces motivic periods — specifically, multiple zeta values $\zeta(3), \zeta(5), \zeta(7), \ldots$ — as genuine physical content of loop integrals, not mere regularization artifacts. We observe that this "reification of ζ at positive integers" has a natural counterpart: the vacuum zero-point energy, regularized via ζ at *negative* integers ($\zeta(-1), \zeta(-3), \zeta(-5), \ldots$, equivalently Bernoulli numbers), also carries physical content. The Riemann functional equation $\zeta(s) \leftrightarrow \zeta(1-s)$ connects these two sides. We propose that the *full* physical content of the ζ function comprises both wings: the **perturbative wing** (Kreimer–Connes, ζ at positive integers, loop integrals) and the **non-perturbative wing** (vacuum cosmology, ζ at negative integers, boundary-condition-dependent zero-point sums). As evidence, we show: (i) the fine structure constant α^{-1} decomposes into an algebraic leading part from $\zeta(-1), \zeta(-3)$ (Bernoulli) plus transcendental corrections from $\zeta(6), \zeta(7)$ (periods) — precisely the structure predicted by the Deligne–Beilinson period conjectures; (ii) the dark energy density $\Omega_\Lambda = 16|\zeta(-1)|\zeta(2)/\pi = 2\pi/9$ combines both wings in a single observable, derivable independently from Wheeler–DeWitt quantum cosmology with temporal Dirichlet–Neumann boundaries; (iii) the Euler product truncation at $p = 2$ ($\zeta \to \zeta_{\neg 2}$), corresponding to DN boundary conditions, is a non-perturbative selection rule invisible to perturbation theory. This framework positions vacuum cosmology as the natural complement — not competitor — to Kreimer–Connes perturbative renormalization, connected by the functional equation.

1 Introduction

1.1 Kreimer–Connes: ζ values are real in perturbative QFT

The work of Kreimer [**?**] and Connes–Kreimer [**?**, **?**, **?**] established that renormalization in perturbative quantum field theory is not an ad hoc subtraction procedure but a precise algebraic operation: the Birkhoff decomposition of characters on a Hopf algebra of Feynman graphs. A key consequence is that the *finite* (renormalized) parts of Feynman integrals evaluate to **motivic periods** — algebraic combinations of multiple zeta values (MZVs) and polylogarithms [**?**, **?**]:

$$\text{Renormalized Feynman integral} \in \mathbb{Q}[\zeta(3), \zeta(5), \zeta(7), \ldots, \mathrm{Li}_n(\tfrac{1}{2}), \ln 2, \pi^2]. \tag{1}$$

For example, the two-loop QED contribution to the electron anomalous magnetic moment contains $\frac{3}{4}\zeta(3)$ [**?**]. The three-loop involves $\zeta(5)$. Higher loops bring $\zeta(7)$ and MZVs of increasing depth.

Connes and Marcolli [**?**] further showed that a *cosmic Galois group* acts on these periods, giving the arithmetic content of perturbative QFT a Galois-theoretic structure analogous to the motivic Galois group of algebraic geometry.

The philosophical content of this program is profound: ζ **values at positive integers are not regularization artifacts but genuine physical invariants of quantum field theory.** They are periods of algebraic varieties (complements of graph hypersurfaces), and their appearance in physical observables (anomalous dimensions, β-functions, $g-2$) is mathematically inevitable.

1.2 The missing wing

The Kreimer–Connes framework operates within perturbation theory. It addresses the arithmetic of *loop integrals* — fluctuations *around* a background vacuum. It does not address the vacuum itself.

However, the vacuum also has arithmetic content. The zero-point energy of a quantum field in a cavity of size a is formally

$$E_0 = \frac{\hbar}{2} \sum_n \omega_n \tag{2}$$

which diverges but, via ζ-regularization, yields a finite physical quantity — the Casimir energy. The regularization involves ζ at *negative integers*:

$$\zeta(-1) = -\frac{1}{12} = -\frac{B_2}{2}, \quad \zeta(-3) = \frac{1}{120} = -\frac{B_4}{4}, \quad \ldots \tag{3}$$

These are *Bernoulli numbers* — rational, algebraic, on the opposite side of the ζ function from the periods $\zeta(3), \zeta(5), \ldots$ that Kreimer–Connes handles.

The question of this paper is:

> *If Kreimer–Connes establishes that ζ at positive integers is physically real (in perturbative QFT), is ζ at negative integers also physically real (in non-perturbative vacuum physics)?*

We argue **yes**, and show that the Riemann functional equation provides the bridge.

1.3 The functional equation as a physical duality

The Riemann functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin\!\left(\tfrac{\pi s}{2}\right) \Gamma(1-s)\, \zeta(1-s) \tag{4}$$

relates $\zeta(s)$ to $\zeta(1-s)$. At $s = 2$ and $s = -1$:

$$\zeta(2) = \frac{\pi^2}{6} \quad \longleftrightarrow \quad \zeta(-1) = -\frac{1}{12}. \tag{5}$$

The left side is a transcendental period; the right side is a rational Bernoulli value. They are "dual" under the functional equation.

We propose that this mathematical duality has a *physical* counterpart:

	Perturbative wing	**Non-perturbative wing**
ζ arguments	Positive integers	Negative integers
Values	Periods (π^{2k}, MZVs)	Bernoulli ($B_{2k}/(2k)$)
Physics	Loop integrals	Vacuum zero-point
Framework	Kreimer–Connes	Wheeler–DeWitt DN
Observables	a_e, α running	Ω_Λ, Casimir
Math nature	Transcendental	Algebraic (rational)

The functional equation (**??**) connects these wings, and we will show that specific physical observables — notably α^{-1} and Ω_Λ — involve *both* wings simultaneously.

2 The two wings in detail

2.1 Perturbative wing: Kreimer–Connes

We briefly review the Kreimer–Connes results relevant to our argument.

Hopf algebra of Feynman graphs. Let $\mathcal{H}$ be the commutative Hopf algebra generated by one-particle irreducible Feynman graphs, with coproduct encoding subdivergence structure [**?**]. A *character* $\phi : \mathcal{H} \to \mathcal{A}$ (where $\mathcal{A}$ is an algebra of Laurent series in the regularization parameter ε) encodes the evaluation of each graph.

Birkhoff decomposition. The character ϕ admits a unique factorization $\phi = \phi_-^{-1} \star \phi_+$, where ϕ_- contains the poles (counterterms) and ϕ_+ is holomorphic (renormalized values) [**?**]. This is the Bogoliubov–Parasiuk–Hepp–Zimmermann (BPHZ) renormalization in algebraic form.

Periods. The renormalized values $\phi_+(\Gamma)$ for primitive graphs Γ evaluate to periods of mixed Tate motives over $\mathbb{Z}$ [**?**]. Concretely, they are $\mathbb{Q}$-linear combinations of products of $\zeta(n)$ ($n \geq 2$ integer) and $\ln 2$. For massless propagators, only odd ζ-values $(\zeta(3), \zeta(5), \ldots)$ appear at the single-zeta level.

Physical consequence. The QED electron $g-2$, computed to five loops, contains $\zeta(3), \zeta(5)$ at 2–3 loops, and deeper MZVs at higher loops [**?**, **?**]. These are *measured* through the electron $g-2$ experiment, making them empirically verified physical quantities.

2.2 Non-perturbative wing: vacuum zero-point

The vacuum zero-point energy (**??**) depends on *boundary conditions* imposed on the field modes:

- **Dirichlet–Dirichlet (DD)** or **Neumann–Neumann (NN)**: All-integer modes $n = 1, 2, 3, \ldots$. Regularized sum $\sum n^{-s}|_{s=-1} = \zeta(-1) = -1/12$.
- **Dirichlet–Neumann (DN)**: Odd-integer modes $n = 1, 3, 5, \ldots$. Regularized sum $\sum_{\text{odd}} n^{-s}|_{s=-1} = \zeta_{\neg 2}(-1) = +1/12$.

where

$$\zeta_{\neg 2}(s) \equiv (1 - 2^{-s})\zeta(s) \tag{6}$$

is the ζ function with the $p = 2$ Euler factor removed.

Key difference from Kreimer–Connes: this is *not* a loop integral. It is the sum over all modes of the *background vacuum*, without perturbative expansion. The ζ values at negative integers $(\zeta(-1), \zeta(-3), \ldots)$ emerge not from Feynman diagrams but from the *mode spectrum* of the vacuum itself.

Physical consequence. The 3D Casimir effect gives a measurable force $\propto \zeta(-3) = 1/120$ [**?**], confirmed experimentally [**?**]. Boyer's DN Casimir prediction involves $\zeta_{\neg 2}(-3) = -7/120$ [**?**], not yet experimentally verified.

2.3 Why Kreimer–Connes cannot see the non-perturbative wing

Perturbation theory expands around a *fixed* vacuum. The vacuum itself is the zeroth-order background. Its zero-point energy is subtracted (normal ordering) or absorbed into the cosmological constant. Kreimer–Connes, operating within perturbation theory, systematically handles all loop corrections *above* the vacuum, but the vacuum energy itself lies *outside* the Hopf algebra of Feynman graphs.

This is precisely why perturbative QFT cannot solve the cosmological constant problem: Λ is a non-perturbative vacuum property, not a loop correction.

3 Evidence: α^{-1} spans both wings

3.1 Decomposition

We have previously reported [?] the arithmetic formula

$$\alpha^{-1} \approx \underbrace{\frac{2}{|B_2|} + \frac{4}{|B_4|}}_{\text{algebraic (Bernoulli)}} + \underbrace{g \cdot 132 + 4(\zeta(6) - \zeta(7))}_{\text{transcendental (periods)}} \tag{7}$$

where $2/|B_2| = 12$, $4/|B_4| = 120$, $g \approx 0.038$, and $4(\zeta(6) - \zeta(7)) \approx 0.036$.

We now observe that this decomposition has a natural interpretation in terms of the two wings:

- **Leading (algebraic) part**: $12 + 120 = 132$. These are $1/|\zeta(-1)| + 1/|\zeta(-3)|$ — values at *negative* integers, the non-perturbative wing. They represent the "vacuum contribution" to α^{-1}.
- **Correction (transcendental) part**: $g \cdot 132$ and $\zeta(6) - \zeta(7)$. These involve periods at *positive* integers, the perturbative wing. They represent the "loop corrections" to α^{-1}.

3.2 Deligne–Beilinson structure

The Deligne period conjecture [?] and Beilinson conjectures [?] predict that critical L-values of motives factor as

$$L^*(M, n) \sim c^+(M) \cdot \Omega(M) \tag{8}$$

where c^+ is algebraic and Ω is a period.

Formula (**??**) has precisely this form:

$$\alpha^{-1} = \underbrace{132}_{\text{algebraic } c^+} + \underbrace{\text{transcendental corrections}}_{\text{period } \Omega} \tag{9}$$

If this identification holds, α^{-1} is a physical instance of the Beilinson conjectures, with the algebraic part from the non-perturbative wing and the period part from the perturbative wing.

4 Evidence: Ω_Λ combines both wings

4.1 The functional-equation dual form

We have shown [?] that the dark energy density parameter

$$\Omega_\Lambda = \frac{2\pi}{9} \tag{10}$$

admits the representation

$$\Omega_\Lambda = \frac{16\,|\zeta(-1)|\,\zeta(2)}{\pi} \tag{11}$$

which explicitly combines:

- $\zeta(-1) = -1/12$: non-perturbative wing (Bernoulli, algebraic)
- $\zeta(2) = \pi^2/6$: perturbative wing (period, transcendental)
- $1/\pi$: normalization (from Einstein's equation $8\pi G$)

Ω_Λ lives on the bridge between the two wings.

4.2 Wheeler–DeWitt derivation

The derivation proceeds via a non-perturbative argument [**?**]:

1. The Wheeler–DeWitt wave function $\Psi(a)$ satisfies temporal Dirichlet–Neumann boundary conditions (Dirichlet at Big Bang, Neumann at de Sitter asymptotic).
2. The 1D DN zero-point energy regularizes to $\zeta_{\neg 2}(-1) = +1/12$ (non-perturbative wing, with Euler $p = 2$ truncation).
3. The Cohen–Kaplan–Nelson holographic bound [**?**] provides $\rho_\Lambda \leq M_P^2/\ell_H^2$ (the bridge between UV and IR).
4. Saturation with coefficient 1/12 yields $\Omega_\Lambda = 2\pi/9$.

Steps 1–2 are purely non-perturbative (no loops). Step 3 involves the holographic principle (UV/IR mixing). The final result (**??**) involves $\zeta(2)$ (perturbative wing) via the 8π of Einstein's equation.

4.3 Why perturbation theory cannot produce Ω_Λ

This explains why the cosmological constant problem has resisted perturbative solution for 40+ years [**?**]. The vacuum energy ρ_Λ is a non-perturbative quantity (ζ at negative integers) that perturbation theory (ζ at positive integers) cannot access. Kreimer–Connes, being perturbative, is structurally unable to predict Ω_Λ. The non-perturbative wing is required.

5 The $p = 2$ Euler truncation as non-perturbative selection

5.1 DN boundary and $\zeta_{\neg 2}$

The Euler product of ζ is

$$\zeta(s) = \prod_{p \text{ prime}} (1 - p^{-s})^{-1}. \tag{12}$$

Removing the $p = 2$ factor gives

$$\zeta_{\neg 2}(s) = (1 - 2^{-s})\zeta(s) = \prod_{p \geq 3} (1 - p^{-s})^{-1}. \tag{13}$$

Physically, $\zeta_{\neg 2}$ arises from DN boundary conditions (odd modes only). The removal of the $p = 2$ Euler factor corresponds to the *elimination of even harmonics* by the mixed boundary.

5.2 Invisibility to perturbation theory

In perturbative QFT, all modes contribute to loop integrals. There is no mechanism to select odd modes only. The Euler product truncation at $p = 2$ is a *boundary-condition effect* on the vacuum, not a loop-level phenomenon.

Proposition 1 (Non-perturbative character of $\zeta_{\neg 2}$)**.** *The transition $\zeta \to \zeta_{\neg 2}$ (equivalently, DD $\to$ DN boundary) is invisible to perturbation theory. It is a non-perturbative selection rule operating at the vacuum level.*

This is why Kreimer–Connes never encounters $\zeta_{\neg 2}$. The perturbative wing uses the *full* ζ (all primes). The non-perturbative wing uses $\zeta_{\neg 2}$ (odd primes only) when DN boundaries are present.

5.3 Sign consequence

At $s = -1$:

$$\zeta(-1) = -\frac{1}{12} \quad \text{(NN/DD: attractive vacuum)} \tag{14}$$

$$\zeta_{\neg 2}(-1) = +\frac{1}{12} \quad \text{(DN: repulsive vacuum)} \tag{15}$$

The sign flip is physically decisive: positive vacuum energy ($\rho_\Lambda > 0$) drives cosmic acceleration. Only DN boundary conditions produce the correct sign. This is a non-perturbative datum that perturbation theory has no access to.

6 Unified picture

6.1 The zeta building

We propose the metaphor of a "zeta building" with two wings:

$$\underbrace{\zeta(-5), \zeta(-3), \zeta(-1)}_{\text{Non-perturbative wing}} \quad [\text{functional equation}]\zeta(s) \leftrightarrow \zeta(1-s) \quad \underbrace{\zeta(2), \zeta(3), \zeta(5), \zeta(7)}_{\text{Perturbative wing}} \tag{16}$$

- **Left (non-perturbative) wing**: Bernoulli numbers, vacuum zero-point, Casimir effect, Ω_Λ, boundary-condition dependence. Algebraic values. Accessible via Wheeler–DeWitt quantum cosmology.
- **Right (perturbative) wing**: Periods, MZVs, loop integrals, anomalous dimensions, $g-2$. Transcendental values. Accessible via Kreimer–Connes Hopf algebra.
- **Connecting hallway**: The Riemann functional equation. Physical observables like α^{-1} and Ω_Λ span both wings.

6.2 Physical observables and their wing structure

Observable	Left wing	Right wing	Bridge
a_e (electron $g-2$)	—	$\zeta(3), \zeta(5)$	KC only
Casimir (DD)	$\zeta(-3)$	—	Left only
Casimir (DN)	$\zeta_{\neg 2}(-3)$	—	Left only
Ω_Λ	$\zeta_{\neg 2}(-1)$	$\zeta(2)$ via $8\pi G$	Both
α^{-1}	$1/\lvert\zeta(-1)\rvert, 1/\lvert\zeta(-3)\rvert$	$\zeta(6), \zeta(7)$	Both

Some observables live purely in one wing; others span both. The spanning observables (α^{-1}, Ω_Λ) are the most physically fundamental, as they require the full ζ building for their description.

7 Relation to existing programs

7.1 Kreimer–Connes: the perturbative wing is established

The right wing is mathematically rigorous and experimentally verified (electron $g-2$ to $\sim 10^{-13}$ precision). Our proposal does not modify or challenge Kreimer–Connes. It *extends* the same philosophy ("ζ values are physically real") to the non-perturbative side.

7.2 Casimir physics: the left wing is partially established

The Casimir effect for DD boundaries is experimentally confirmed [?]. The $\zeta(-3) = 1/120$ coefficient appears in the measured force. Boyer's DN prediction ($\zeta_{\neg 2}(-3) = -7/120$) is not yet experimentally tested.

Our proposal extends this to cosmological scales via the Wheeler–DeWitt temporal DN argument.

7.3 Deligne–Beilinson: the bridge is conjectured

The Deligne and Beilinson conjectures [?, ?] predict that L-values factor into algebraic and transcendental (period) parts. Our observation that α^{-1} and Ω_Λ have precisely this structure is a candidate physical realization.

7.4 Moonshine: a parallel

Monstrous Moonshine [?] demonstrated that modular forms (j-function) encode genuine physics (Monster VOA, 2D CFT). Mathieu Moonshine [?] extended this to K3 surfaces. Our proposal that ζ special values encode 4D physical constants is a parallel "ζ-moonshine" — not via a sporadic group, but via the functional equation duality of the ζ function itself.

8 Predictions

The unified framework makes specific testable predictions:

1. $\Omega_\Lambda = 2\pi/9$: Testable at 0.5% by Euclid/DESI/LSST (2025–2028). This is a purely non-perturbative-wing prediction, inaccessible to perturbation theory.

2. **DN Casimir (Boyer)**: The 3D DN Casimir coefficient $\zeta_{\neg 2}(-3) = -7/120$ (repulsive) can be tested in laboratory experiments with mixed-boundary acoustic or electromagnetic resonators.

3. α^{-1} **decomposition**: The split of α^{-1} into algebraic (132) and transcendental corrections can be compared against high-precision QED calculations. If the split is exact, higher-loop terms should organize into period-valued corrections to the Bernoulli leading part.

4. **Muon and tau** $g-2$: The identification $\Delta a_\mu \approx 252 = 6/|B_6|$ (non-perturbative wing) and the Ramanujan discriminant $\tau(3) = 252$ suggest that generational anomalies may bridge both wings. Prediction: $\Delta a_\tau \propto \tau(5) = 4830$.

9 Limitations

1. The "two-wing" picture is a *proposal*, not a theorem. The functional equation is a mathematical identity; its physical interpretation as a duality between perturbative and non-perturbative physics is conjectural.

2. The Wheeler–DeWitt derivation of Ω_Λ assumes specific boundary conditions (temporal DN) and CKN bound saturation. These assumptions are physically motivated but not derived from first principles.

3. The decomposition of α^{-1} into algebraic + period parts is observed post hoc. A rigorous derivation via Kreimer–Connes Hopf algebra is future work.

4. The Euler $p = 2$ truncation (DN boundary) is presented as a non-perturbative selection rule, but its fundamental origin (why $p = 2$? why DN?) remains open. Candidates include spin structure ($K_1(\mathbb{Z}) = \mathbb{Z}/2$) and time-reversal asymmetry.

5. This framework does not address fermion masses, Yukawa couplings, or CKM mixing — these may require additional structure beyond the ζ building.

10 Discussion

The central thesis of this paper is that the arithmetic content of quantum physics is organized by the Riemann ζ function, with perturbative physics on one side of the functional equation and non-perturbative vacuum physics on the other.

Kreimer–Connes brilliantly established the perturbative side. They showed that ζ values at positive integers are not regularization artifacts but genuine motivic periods with Galois-theoretic structure. Our contribution is to extend this philosophy to the other side: ζ values at negative integers (Bernoulli numbers) appearing in vacuum energy regularization are not computational conveniences but genuine physical data about the boundary-condition structure of the vacuum.

If this picture is correct, the cosmological constant problem is not a failure of quantum field theory per se, but a consequence of *looking only at the perturbative wing.* The full ζ building, spanning both wings via the functional equation, naturally produces the observed small positive Λ from the non-perturbative wing ($\zeta_{\neg 2}(-1) = +1/12$) combined with the holographic bridge (CKN bound).

The beauty of this picture lies in its economy: no new fields, no extra dimensions, no multiverse. Only the recognition that the ζ function's functional equation is not merely a mathematical symmetry but a *physical duality* between the perturbative and non-perturbative sectors of quantum physics.

11 Conclusion

We have proposed that the physical content of the Riemann ζ function comprises two complementary wings:

- The **perturbative wing** (ζ at positive integers: periods, MZVs, loop integrals), established by Kreimer–Connes.
- The **non-perturbative wing** (ζ at negative integers: Bernoulli numbers, vacuum zero-point, boundary conditions), proposed by this work via Wheeler–DeWitt quantum cosmology.

The Riemann functional equation connects these wings, and specific physical observables (α^{-1}, Ω_Λ) span both. The framework provides:

1. A parameter-free prediction $\Omega_\Lambda = 2\pi/9$ (2% agreement with Planck 2018).
2. A Deligne–Beilinson-type decomposition of α^{-1} into algebraic and period parts.
3. An explanation of why the cosmological constant problem resists perturbative solution (it lives in the non-perturbative wing).
4. A non-perturbative selection rule ($\zeta \to \zeta_{\neg 2}$, DN boundary) invisible to loop expansion.

The proposal positions vacuum cosmology as a natural complement to Kreimer–Connes perturbative renormalization. Together, they may constitute the two halves of a complete arithmetic description of quantum physics — united by the functional equation of the Riemann ζ function.

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